

Deep Hedging under Transaction Costs: A Reinforcement Learning Study

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Abstract—This paper investigates whether a Proximal Policy Optimization (PPO) reinforcement learning agent can replace the Black-Scholes delta-hedging strategy for a vanilla European call option under discrete rebalancing and transaction costs. The hedging problem is formulated as a Markov Decision Process with a state space and reward function normalized by the strike price, and the agent is trained on a single at-the-money contract using simulated price paths. In the absence of transaction costs, the agent recovers a policy closely resembling the theoretical Black-Scholes delta and generalizes across unseen strike levels. Once transaction costs are introduced, the learned policy trades far less, attaining a higher mean terminal portfolio value at the cost of substantially higher variance. The agent also proves to be relatively robust to volatility mismatch, degrading in a predictable way. These results indicate that reinforcement learning is a viable hedging approach, provided the risk preference is encoded in the objective.

Index Terms—Reinforcement learning, Proximal Policy Optimization, Delta hedging, Transaction costs, Black-Scholes model

I. INTRODUCTION

As of June 2025, the notional amount of outstanding over-the-counter (OTC) derivatives stood at \$846 trillion, a 16% increase over the preceding year [1]. This size expresses the importance of financial derivatives in the financial markets today. Financial institutions use these derivatives extensively, both as tradable products and as instruments for risk management and price stabilization [2]. However, trading and/or holding derivatives in a portfolio is inherently risky. Offsetting the risk of holding these assets, also called hedging, poses a challenge for these institutions, as the price paths of the underlying assets are uncertain and driven by stochastic movements.

Black and Scholes developed a closed-form solution for the hedging problem by replicating the portfolio with some amount of the option's underlying asset. However, this portfolio-replication approach simplifies the environment from reality with several assumptions [3]. This paper addresses two of these: the continuous rebalancing assumption, and the no-transaction-costs assumption. In practice, financial institutions do not rebalance continuously, but rather at discrete steps throughout the option's lifetime. Additionally, in real financial markets there is almost never a situation where transaction costs are not present.

This paper relaxes both assumptions by introducing discrete rebalancing steps and proportional transaction costs when buying or selling the underlying asset. The goal is to use

reinforcement learning, specifically a Proximal Policy Optimization (PPO) agent trained on simulated price paths, to find a hedging strategy under these conditions, and to benchmark this learned strategy directly against the Black-Scholes delta-hedging strategy.

This study contributes to the existing literature by showing that reinforcement learning is able to find a strategy similar to that of the Black-Scholes model in the absence of transaction costs. Additionally, it can be stated that the agent can be trained on one strike level, and that the learned policy generalizes to other strike prices. When transaction costs are introduced, the policy learned by the reinforcement learning agent is very different. Instead of tracking the theoretical hedge, the agent learns a near-static policy, hedging only when the option is likely to expire in the money, and even then only partially. This reduces the transaction costs incurred, but comes at the expense of a substantially higher variance of the final portfolio value. Finally, when evaluated on assets with different volatility regimes than those during training, hedging performance decreases for both the agent and the benchmark, although the agent's error grows comparatively slowly, indicating a relatively robust policy.

The paper is structured as follows: in Section II relevant concepts and literature are explained and reviewed. In Section III the problem is formally defined as a Markov Decision Process (MDP). The reinforcement learning approach and the different experiments are described in Section IV. The results of the experiments are presented in Section V, after which the practical implications, limitations, and possible future work are discussed in Section VI. The paper is concluded and summarized in Section VII.

II. RELATED LITERATURE

In this section, the relevant concepts will be explained, along with the relevant papers pointing out earlier research directions. The section is structured as follows: First the concepts of hedging, financial derivatives, and the Black-Scholes model are introduced. Next, the Markov decision process structure is explained and motivated, after which Reinforcement Learning is introduced. In the fourth subsection some examples and applications of Reinforcement Learning for hedging are discussed, and finally the research aim of this paper is stated.

A. Hedging & the Black-Scholes Framework

A financial derivative is a contract or product that derives its value from the price of another asset. There are many types of derivatives, but one of the most basic, yet key to the hedging problem addressed here, is the European-style option. A European option gives its holder the right, but not the obligation, to buy (call) or sell (put) the underlying asset at an agreed strike price K , with this right only exercisable at a fixed maturity date T . This is unlike American-style options, which permit early exercise at any point up to T . This restriction to a single exercise date makes the European option analytically solvable, allowing it to be priced in closed form, which is precisely what makes it the natural starting point for researching the hedging problem addressed in this paper.

One of the key formulas in derivative pricing is the Black-Scholes formula [3]. The key insight here is that a derivative is priced by constructing a portfolio Π of some amount of stocks, denoted by the delta Δ , that offsets the risk of selling that particular derivative. This offsetting of the risk is called hedging. In the case of the Black-Scholes model, the derivative hedged is a European option, where the price dynamics of the underlying asset are assumed to follow Geometric Brownian Motion (GBM) [4]. The payoff of a European call option is defined by the following formula:

$$V(S(T), T) = \max(S(T) - K, 0) \quad (1)$$

where K is the strike price and T the maturity. $S(t)$ follows GBM dynamics defined as:

$$dS(t) = \mu S(t) dt + \sigma S(t) dW(t) \quad (2)$$

where μ is the drift, σ the volatility, and $W(t)$ a standard Wiener process.

To hedge a written option, a replicating portfolio is constructed consisting of the option and an offsetting position of Δ shares of the underlying asset:

$$\Pi = V - \Delta S \quad (3)$$

The position in the underlying is financed through a bank account earning the risk-free rate. This means purchases of the asset are funded by borrowing, sales are deposited, and the portfolio is *self-financing* in the sense that no external capital is added or withdrawn after construction [5]. In continuous time this financing account is typically left implicit in the notation of Equation 3; in the discrete-time hedging problem studied in this paper, however, it must be modeled explicitly.

Requiring this portfolio Π to be risk-free, and thus earn the risk-free rate under a no-arbitrage argument, yields the following partial differential equation (PDE), better known as the Black-Scholes equation [3]:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0 \quad (4)$$

where r is the risk-free rate. Notably, the drift μ of the underlying asset, present in Equation 2, does not appear in Equation 4. This reflects a key property of the delta-hedging

argument: continuous rebalancing according to Δ causes the random fluctuations of the hedged portfolio to exactly offset those of the option, regardless of the asset's actual drift. As a result, both the option price and the hedge ratio Δ are independent of μ . Solving Equation 4 for a European call under the boundary condition given by Equation 1 yields the closed-form Black-Scholes price, and correspondingly the closed-form hedge ratio:

$$\Delta = N(d_1), \quad d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}} \quad (5)$$

where $N(\cdot)$ denotes the standard normal cumulative distribution function.

There are some critical limitations to the Black-Scholes model, as it assumes normally distributed returns, continuous rebalancing, no transaction costs, and constant volatility. These are all unrealistic assumptions, which complicates the direct use of the Black-Scholes model for rebalancing the portfolio [6].

The first hedging parameter is the delta Δ , given in Equation 5. This is the fraction of stock to be held to offset the risk, and can also be seen as the sensitivity of the option price V to a change in the asset price S . The second hedging parameter is called Gamma, defined as

$$\Gamma = \frac{\partial^2 V}{\partial S^2} = \frac{\partial \Delta}{\partial S}, \quad (6)$$

which measures the sensitivity of the delta to changes in the underlying asset price. Both parameters must be neutralized to fully remove the portfolio's risk exposure.

B. Sequential Decision Making

As the price of an asset changes, so do the delta and Gamma values. Adjusting the portfolio such that there is no risk exposure requires frequent rebalancing according to the hedging parameters. Because the price movements of an asset are uncertain, and at every new observed price there is a new decision to make with regard to the rebalancing, hedging can be seen as a sequential decision problem [7].

Broadly speaking, the sequential structure is to observe the price of an asset, rebalance the portfolio, and incur the associated transaction costs. This sequential structure can thus be modeled as a Markov Decision Process (MDP), which has been done extensively in past literature [8], [9]. Several solutions have been proposed, using dynamic programming solutions, stochastic control, and linear programming for example [10]. However, these solutions all suffer from the curse of dimensionality, prohibiting the use of very complex state and/or action spaces [11]. In this paper, the rebalancing problem will also be modeled using an MDP approach, which will be formally defined in Section III.

An MDP consists of a state space, that defines the current state of a system [12]. Additionally, the action space defines the possible actions that can be taken in a specific state. The dynamics for moving to the next state are governed by the transition functions, where the agent is rewarded for the action

it took based on pre-determined rules. The final policy that maximizes or minimizes the specified total rewards over the horizon can be formulated using the Bellman Equation. It is precisely this curse of dimensionality — where classical dynamic programming and stochastic control solutions become intractable as the state and action space grow — that motivates the use of Reinforcement Learning, discussed next.

C. Reinforcement Learning

As described in the previous section, classical solutions to the hedging MDP become intractable as the state and action space grow. Reinforcement Learning (RL) offers a way around this, by allowing an agent to learn a policy through direct interaction with an environment, rather than requiring an explicit solution of the underlying dynamic programming problem [13]. At every timestep, the agent observes a state s_t , takes an action a_t , and receives a reward r_t , after which the environment transitions to a new state s_{t+1} . Through repeated interaction, the agent updates its policy π to maximize the expected cumulative reward, without requiring a model of the environment’s transition dynamics to be known in advance.

RL algorithms can broadly be divided into value-based and policy-based methods. Value-based methods, such as Q-learning [14], learn an action-value function $Q(s, a)$ that estimates the expected return of taking action a in state s , from which a policy is derived by acting greedily with respect to Q . The introduction of deep neural networks as function approximators for $Q(s, a)$, referred to as Deep Q-Networks (DQN), enabled value-based methods to scale to high-dimensional state spaces [15], with Double DQN [16] later addressing the overestimation bias present in the original algorithm. Policy-based methods instead parameterize the policy π_θ directly and optimize its parameters via gradient ascent on the expected return, without necessarily learning a value function as an intermediate step. Proximal Policy Optimization (PPO) [17] is a widely used policy-based method, which constrains each policy update to remain close to the previous policy, improving training stability compared to earlier policy gradient methods.

D. Reinforcement Learning for Hedging Derivatives

The problem of hedging and rebalancing the portfolio using RL has been studied extensively. In the paper of Cao et al. [8], the price of the asset is modeled using GBM and a stochastic volatility process. A reinforcement learning model was then applied to approximate different Q-value functions, from which the optimal policy is obtained by acting greedily with respect to the Q-values.

The study by Mikkilä and Kanninen [18] uses an actor-critic reinforcement learning model that enables continuous action spaces. This is relevant to the problem of rebalancing as both the delta and Gamma parameters are continuous-valued. Discretizing these values, as required by value-based approaches, therefore loses accuracy. This particular study also included a Heston model for generating asset prices, meaning volatility is stochastic.

The study by Buehler et al. [19] introduced the “Deep Hedging” framework, in which a neural network policy is trained directly to minimize a risk measure, such as Conditional Value at Risk (CVaR), rather than learning a value function. This formulation allows the framework to incorporate market frictions such as transaction costs and liquidity constraints directly into the training objective, and has since become a common baseline against which subsequent RL-based hedging approaches, including those discussed above, are compared.

E. Problem Statement

Earlier work on Reinforcement Learning for option hedging spans a range of algorithmic approaches, from value-based methods that discretize the hedging action [8], to continuous-action actor-critic methods evaluated under stochastic volatility [18], to risk-sensitive frameworks that directly incorporate market frictions into the training objective [19]. However, Proximal Policy Optimization (PPO) [17] has not been directly benchmarked against the closed-form Black-Scholes delta-hedging strategy for a vanilla European call option under realistic market frictions such as transaction costs.

This paper addresses this gap by formulating the delta-hedging problem as a Markov Decision Process and training a PPO agent to learn a hedging policy directly from simulated price paths, without relying on the closed-form Black-Scholes solution. Concretely, the agent is tasked with learning the optimal hedge ratio itself through interaction with the environment, rather than being provided with or trained to imitate the theoretical Black-Scholes delta given in Equation 5. The performance of this learned policy is then directly compared against the Black-Scholes delta-hedging benchmark. This leads to the following main research question and accompanying sub-questions:

Main RQ: Can a PPO-trained reinforcement learning agent be used as a suitable replacement for the Black-Scholes delta-hedging strategy when hedging a vanilla European call option?

RQ1: How does the PPO agent’s hedging performance compare to the Black-Scholes delta-hedging strategy in the absence of transaction costs?

RQ2: How does the relative performance of the PPO agent change as transaction costs are introduced?

RQ3: How does the PPO agent’s hedging performance vary across different levels of moneyness and time to maturity?

RQ4: How does the PPO agent’s hedging performance degrade when tested on price paths with volatility different from that used during training?

RQ5: To what extent does the policy learned by the PPO agent resemble the theoretical Black-Scholes delta given in Equation 5?

III. MARKOV DECISION PROCESS FORMULATION

In this section, the relevant components of the MDP as described in Section II-B are formally defined. For the purpose of this research, we assume that one European call with

maturity T and strike price K is sold on the market. The agent's objective is to construct a portfolio Π using the underlying asset S such that the risk of the short position on the option is hedged. Throughout, $t \in \{0, 1, \dots, N\}$ indexes the discrete rebalancing dates, with date t corresponding to calendar time $t \Delta t$ and the final index N to the maturity T .

A. State Space

The state space consists of three components known at any time t before the action is decided. The stock price $S(t)$ is known, but to indicate the “moneyness” of the option contract, the first feature for the state space will be the stock price divided by the strike price, $S(t)/K$. Additionally, the time to maturity should be included in the state space to give the decision maker a sense of time and maturity. This gives the second state space component t/N , the fraction of the option's life elapsed. The third and last component of the state space is the current fraction of shares that is held in the portfolio from earlier transactions, defined as $\rho(t-1)$. As in the Black-Scholes model, the risk-free rate r and volatility σ are treated as fixed and known constants of the environment for a given episode, rather than time-varying quantities, and are therefore not included as state variables. The full state space is given by:

$$s_t = \left(\frac{S(t)}{K}, \frac{t}{N}, \rho(t-1) \right) \quad (7)$$

Although the environment additionally tracks a bank account $B(t)$, introduced in Section III-C, it is not included as a state variable: given the trading history, it evolves deterministically and enters the objective only through the terminal reward, so the state in Eq. (7) remains sufficient for optimal decision-making.

B. Action Space

For the action space, the agent chooses when, and how much, to rebalance the portfolio, with the aim of hedging the risk of selling the European call option. From the literature in Section II-A we know that some amount of stock, the delta $\Delta(t)$, will offset the risk of the portfolio at any time t . The decision here is thus when, and how much, of asset S to buy or sell relative to the current holdings $\rho(t-1)$. The action a_t represents the change to be added to or subtracted from the agent's currently held position. The action space is defined as:

$$a_t \in \mathcal{A} = [-1, 1] \quad (8)$$

Because $\rho(t) \in [0, 1]$, consistent with the theoretical bound on the Black-Scholes delta, the action at any timestep is further constrained by the agent's current holdings:

$$-\rho(t-1) \leq a_t \leq 1 - \rho(t-1) \quad (9)$$

This restriction confines the agent to long-only positions in the underlying asset, which ensures the appropriate search space for hedging a short call and excludes short-stock policies by construction.

C. Transition Dynamics

Given the current state and chosen action, the environment transitions to the next state as follows. The held position updates deterministically according to the agent's action:

$$\rho(t) = \rho(t-1) + a_t \quad (10)$$

The time to maturity also decreases deterministically as $t \rightarrow t+1$, while the stock price moving from $S(t) \rightarrow S(t+1)$ is governed by a stochastic process, the specific form of which is described in Section IV.

In addition to the held position, the agent maintains a bank account $B(t)$ that finances all trading. The purchase or sale of shares, together with the associated transaction cost, is settled through this account, which accrues interest at the risk-free rate between rebalancing dates:

$$B(t) = B(t-1) e^{r \Delta t} - a_t \cdot S(t) - c(t) \quad (11)$$

where $\Delta t = T/N$ denotes the time between the N discrete rebalancing dates, and $c(t)$ denotes the transaction cost, defined in Section III-D. Interest accrues on the incoming balance over the preceding period, after which the trade at date t is settled at the prevailing price $S(t)$. The bank account is initialized with the Black-Scholes premium of the option, $B(0) = V_{BS}(S(0), 0)$, reflecting that the hedger receives the option premium at inception and must replicate the payoff from this initial capital. The hedging strategy is thus self-financing: no external funds are added or withdrawn during the life of the option. This is consistent with the replicating-portfolio construction underlying the Black-Scholes framework.

D. Reward Function

When buying or selling assets, there is some transaction cost to be paid. Let β be some constant transaction cost percentage, proportional to the notional amount traded. The transaction cost at any date $t < N$, denoted $c(t)$, is given by:

$$c(t) = \beta \cdot |a_t| \cdot S(t) \quad (12)$$

where $|a_t|$ ensures a positive cost is incurred regardless of whether the action corresponds to buying or selling the asset. As transaction costs are settled through the bank account in Eq. (11), they implicitly compound at the risk-free rate over the remaining life of the option.

At maturity T , the call option expires. This means that as the seller of this contract, we are obliged to pay the final payoff $V(S(T), T) = \max(S(T) - K, 0)$. To cover this payoff, we sell the amount of the asset obtained over time, $\rho(T)$. Unlike the rebalancing decisions taken at dates $t < N$, this final liquidation is not an action chosen by the agent, but a forced sale of the entire remaining position, and thus incurs its own transaction cost:

$$c(T) = \beta \cdot \rho(T) \cdot S(T) \quad (13)$$

The terminal value of the portfolio is the sum of the bank account and the proceeds of the final liquidation, less the option payoff owed:

$$\Pi(T) = B(T) + \rho(T) \cdot S(T) - c(T) - V(S(T), T) \quad (14)$$

Since the bank account is initialized with the option premium, $\Pi(T)$ measures the replication error of the premium-funded hedge: a perfect hedge reproduces the payoff exactly from the initial premium, corresponding to $\Pi(T) = 0$; any deviation from zero, in either direction, represents a hedging error.

The reward at each date $t < N$ is defined as the negative transaction cost, normalized by the strike price K :

$$r_t = -\frac{c(t)}{K}, \quad t < N \quad (15)$$

At maturity, the agent receives a terminal reward penalizing the absolute deviation of the portfolio value from zero, normalized by K :

$$r_N = -\frac{|\Pi(T)|}{K} \quad (16)$$

E. Objective Function

The objective is to find a policy π_θ that maximizes the expected cumulative reward over the life of the option, which amounts to minimizing the absolute hedging error at maturity while minimizing the transaction costs incurred through rebalancing. As the episode has a finite horizon, no discounting is applied ($\gamma = 1$), and the objective function is given by:

$$\max_{\theta} \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{N-1} r_t + r_N \right] = \max_{\theta} \mathbb{E}_{\pi_\theta} \left[-\frac{|\Pi(T)|}{K} - \sum_{t=0}^{N-1} \frac{c(t)}{K} \right] \quad (17)$$

where the expectation is taken over the stochastic price paths of the underlying asset and the agent's policy.

IV. METHODOLOGY

In this section, the overall structure of the experiments and the model of choice will be motivated. First, the environment parameters are described, including the stochastic process for $S(t)$. Next, the motivation for using a PPO model is explained, along with the hyperparameters, model selection criteria, and network architecture. Finally, the design of the experiments performed is explained and related to the research questions stated in Section II-E.

A. Simulation Environment

For simulating the asset price $S(t)$, GBM will be used, as it is the standard benchmark model in comparable studies [4], [8]. The dynamics for $S(t)$ are given in Equation 2 as a Stochastic Differential Equation (SDE). However, using Itô's lemma we can derive the analytic solution [5]. The analytic solution is given as:

$$S(t) = S(0) \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W(t) \right] \quad (18)$$

Simulating directly from Equation 18 is exact for GBM, so discretization enters the problem only through the rebalancing frequency, not through the simulated price paths. For choosing

realistic environment parameters the following assumptions are made:

- 1) **Drift and Risk-Free Rate:** The drift of the underlying is assumed equal to the risk-free rate, $\mu = r$, consistent with the drift-independence of the hedging problem discussed in Section II-A. The risk-free rate is set equal to the ten-year Treasury yield of US government bonds as of its most recent close on 2026-07-17. This gives $r = \mu = 0.0455$ [20].
- 2) **Volatility:** For the volatility we look at the paper of Cao et al. [7]. Here a base volatility is assumed of 30%, which gives $\sigma = 0.30$.
- 3) **Time to Maturity:** The duration of the contract is assumed to be one quarter, or 63 trading days. This gives $T = 0.25$ and $N = 63$, where $\Delta t = T/N = 1/252$. This corresponds to daily rebalancing decisions for a 3-month option contract.
- 4) **Asset Price:** In Equation 18 the starting point of the asset path is given by $S(0)$. For the purpose of this research the starting asset price is chosen to be $S(0) = 100$.
- 5) **Strike Price:** Because the state space and reward function (Equation 7 and Equation 15) both generalize for the strike price K , the agent is trained on a single at-the-money contract ($K = 100$) and evaluated on the range $K \in \{80, 85, 90, 95, 100, 105, 110, 115, 120\}$. With this, we test whether the agent is learning a generalizable policy across strike levels unseen during training (Section IV-C1).
- 6) **Transaction Costs:** For the transaction costs β we assume a fixed percentage of the notional volume per trade, as stated in Equation 12. For the purpose of this report, a transaction cost of $\beta = 0.005$, corresponding to 0.5%, is assumed to clearly illustrate the effect on hedging performance. Depending on the experiment, the agent is trained either without transaction costs ($\beta = 0$) or with transaction costs ($\beta = 0.005$), as described in Section IV-C.

For an overview of the different parameters see Table I.

TABLE I: Overview of the simulation environment parameters.

Parameter	Value
Risk-free rate / drift, $r = \mu$	0.0455
Volatility, σ	0.30
Time to maturity, T	0.25 (years)
Number of rebalancing steps, N	63
Timestep size, $\Delta t = T/N$	1/252
Initial asset price, $S(0)$	100
Training strike price, K	100
Evaluation strike price range, K	80–120 (steps of 5)
Transaction cost rate, β	{0, 0.005}

B. Proximal Policy Optimization

Proximal Policy Optimization (PPO) [17] is selected for this study over value-based alternatives such as Double DQN, as the hedging action defined in Equation 8 is continuous, making

a policy-based method a more natural fit than an approach requiring discretization of the action space. Additionally, PPO’s clipped objective constrains the size of each policy update, improving training stability relative to earlier policy gradient methods.

1) *Network Architecture*: The PPO agent consists of separate actor and critic networks, each implemented as a multi-layer perceptron (MLP) with two hidden layers of 64 units and tanh activations, following the default architecture used by the Stable-Baselines3 `MlpPolicy` [21]. The actor network outputs the parameters of a Gaussian distribution over the action a_t , from which actions are sampled during training and clipped to the valid range $\mathcal{A} = [-1, 1]$ defined in Equation 8. The initial standard deviation of this distribution is set via an initial log-standard-deviation of -1.5 , ensuring that early exploration remains largely within the valid action range rather than saturating at its bounds. The critic network outputs a scalar estimate of the state-value function $V(s_t)$, used to compute the advantage estimates during training. Observations are normalized using running mean and standard deviation statistics computed during training; the frozen statistics are applied at evaluation time.

2) *Hyperparameters and Training*: The agent is trained using the hyperparameters summarized in Table II. The discount factor is set to $\gamma = 1.0$ as the episode has a short finite horizon, and a small entropy coefficient is retained to prevent premature policy collapse. Given the scope of this study, an extensive hyperparameter search is not performed. Instead, a small, targeted sensitivity analysis is conducted on the learning rate and clip range ϵ , as these are widely reported to be among the most influential hyperparameters for PPO training stability and performance. The results of this sensitivity analysis are presented in Table III. Learning rates of $\{1 \times 10^{-4}, 3 \times 10^{-4}, 1 \times 10^{-3}\}$ and clip ranges of $\{0.1, 0.2, 0.3\}$ are tested, with the combination yielding the lowest validation hedging error selected for the main experiments. All remaining hyperparameters are held fixed across all runs.

The agent is trained for a total of 1,008,000 environment steps, corresponding to 16,000 episodes given the 63-step episode length, using 8 parallel environments. During training, the policy is evaluated every 10,000 environment steps on 100 held-out validation episodes using the deterministic policy. The checkpoint achieving the best mean validation reward is retained as the final model for each configuration; this guards against late-training policy degradation and is applied uniformly across all configurations. As the agent is trained exclusively on the at-the-money contract ($S(0) = K = 100$, Section IV-A), the hyperparameter sensitivity analysis is likewise conducted using this contract only. This is consistent with the training procedure and avoids using information from the strike levels reserved for evaluating generalization in Section IV-C1.

TABLE II: PPO hyperparameters used for training.

Hyperparameter	Value
Learning rate	$\{1 \times 10^{-4}, 3 \times 10^{-4}, 1 \times 10^{-3}\}$
Clip range, ϵ	$\{0.1, 0.2, 0.3\}$
Discount factor, γ	1.0
GAE λ	0.95
Number of parallel environments	8
Rollout length per environment	252
Batch size	252
Epochs per update	5
Entropy coefficient	0.005
Initial log-standard-deviation	-1.5
Total training steps	1,008,000
Training episodes	16,000

C. Experimental Design

The different experiments that will be performed can be related to the different research questions defined in Section II-E. These experiments will be done with the agent that is trained according to the parameters and method stated in Section IV-B. Additionally, a fixed random seed of 42 is used to ensure reproducible experiments and results, and the best-validation checkpoint of each configuration is used for all evaluations. Here an explanation of these experiments, and how they are related to the specific research questions, is given.

1) *Experiment One*: This experiment aims to answer RQ1: How does the PPO agent’s hedging performance compare to the Black-Scholes delta-hedging strategy in the absence of transaction costs? The agent is trained on a single at-the-money contract, where $S(0) = K = 100$, in an environment without transaction costs, so $\beta = 0$. Training on a single strike allows the resulting policy to be evaluated on strike levels unseen during training, directly testing whether the moneyness-normalized state space defined in Equation 7 is sufficient for the policy to generalize across strikes without being explicitly trained on them. In theory, the agent should converge during training to a policy resembling the theoretical Black-Scholes delta coefficient Δ .

Once trained, the policy is evaluated on $m = 1000$ simulated price paths for each strike level $K \in \{80, 85, 90, 95, 100, 105, 110, 115, 120\}$, with $m = 1000$ chosen to yield a sufficiently precise estimate of the sample mean and standard deviation of the hedging error at each strike. For each strike level, the terminal portfolio value $\Pi(T)$ achieved by the PPO agent is compared against that of the Black-Scholes delta-hedging strategy on the same simulated paths, reported as mean, standard deviation, and mean absolute error, alongside a paired t-test statistic testing whether the difference in hedging error between the two strategies is statistically significant. These results are reported in Table IV. Additionally, the total cumulative difference ϵ_{cum} between the learned position $\rho(t)$ and the theoretical delta $\Delta(t)$, summed over the N rebalancing days, is computed as:

$$\epsilon_{\text{cum}} = \sum_{t=0}^{N-1} |\Delta(t) - \rho(t)| \quad (19)$$

2) *Experiment Two*: This experiment aims to answer RQ2: How does the relative performance of the PPO agent change as transaction costs are introduced? Here the agent is trained on the same at-the-money contract as Experiment One ($S(0) = K = 100$), but in an environment where transaction costs are equal to those determined in Section IV-A, so $\beta = 0.005$. Apart from this different β parameter, the experiment itself and the measures of performance are equal to that of Experiment One (Section IV-C1), including evaluation across the full strike range with $m = 1000$ paths per strike. These results are reported in Table V.

3) *Cross-Experiment Analysis*: The results of Experiment One and Experiment Two are used to answer RQ3: How does the PPO agent’s hedging performance vary across different levels of moneyness and time to maturity? Since both experiments already evaluate ϵ_{cum} across the full range of tested strike levels, results are reported in Table VI, broken down by strike level, to assess whether hedging error varies systematically with moneyness, and whether the policy trained only at $K = 100$ shows degraded performance for strikes further from the training strike.

Additionally, the results of Experiment One and Experiment Two are used to answer RQ5: To what extent does the policy learned by the PPO agent resemble the theoretical Black-Scholes delta given in Equation 5? The learned policy is visualized as a function of moneyness and time-to-maturity and compared directly against the theoretical Black-Scholes delta surface computed from Equation 5. This comparison is made for both the policy trained with and without transaction costs, to assess whether and how the learned policy diverges from the theoretical benchmark once costs are introduced.

To illustrate the hedging behavior of both strategies concretely, three representative simulated price paths are shown, plotting the evolution of $S(t)$, the learned position $\rho(t)$, and the theoretical delta $\Delta(t)$ over the life of the option. This is done for asset paths ending in-the-money (ITM), at-the-money (ATM), and out-of-the-money (OTM).

4) *Experiment Three*: The third experiment aims to answer RQ4: How does the PPO agent’s hedging performance degrade when tested on price paths with volatility different from that used during training? Because the agent is trained on price paths simulated under a fixed base volatility $\sigma = 0.30$ (Section IV-A), evaluating the same trained policy on paths simulated under different volatility levels tests the robustness of the learned policy to a mismatch between training and testing conditions. The policies trained in Experiment One and Experiment Two are not retrained for this experiment. Each is evaluated directly on paths simulated under each alternative volatility level. For the volatility levels tested, values spanning below and above the training volatility are chosen: $\sigma \in \{0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$. For every volatility level, the agent is evaluated on $m = 1000$ paths for each strike level K defined in Section IV-C1.

For each tested volatility level, the mean hedging error of the PPO agent is compared against that of the Black-Scholes delta-hedging strategy, which also hedges under the assumption of

the fixed base volatility $\sigma = 0.30$, with $\Delta(t)$ computed from Equation 5. These results are reported in Table VII, for both the policy trained with and without transaction costs.

V. RESULTS

In this section the results of the experiments described in Section IV-C are presented and analyzed. First, the hyperparameter sensitivity analysis is presented and the configurations used for the main experiments are selected. The hedging performance of the agent trained without transaction costs is then characterized, followed by the agent trained with transaction costs. The two regimes are subsequently compared directly, including the generalization across strike levels. Finally, the robustness of both agents to a volatility mismatch between training and evaluation is assessed.

A. Sensitivity Analysis

Table III reports the best validation reward achieved by each hyperparameter configuration, for both the no-cost and cost regimes. The configuration with learning rate 3×10^{-4} and clip range $\epsilon = 0.2$ achieves the best validation reward in the no-cost regime, and learning rate 1×10^{-3} and clip range $\epsilon = 0.3$ in the cost regime. For the no-cost regime there is a clear difference between the best configuration and those following, which cannot be said for the cost regime, where the differences between configurations are minimal. For the subsequent experiments the configuration for the no-cost regime will also be used for the cost regime, as no clear difference in performance is seen. This avoids introducing a hyperparameter difference as a potential cause of different results in the comparison between the two regimes.

TABLE III: Best validation reward per hyperparameter configuration. The selected configuration is shown in bold.

Learning rate	Clip range ϵ	$\beta = 0$	$\beta = 0.005$
1×10^{-4}	0.1	-0.0169	-0.0609
1×10^{-4}	0.2	-0.0157	-0.0648
1×10^{-4}	0.3	-0.0148	-0.0649
3×10^{-4}	0.1	-0.0147	-0.0599
3×10^{-4}	0.2	-0.0104	-0.0586
3×10^{-4}	0.3	-0.0133	-0.0597
1×10^{-3}	0.1	-0.0127	-0.0600
1×10^{-3}	0.2	-0.0165	-0.0594
1×10^{-3}	0.3	-0.0132	-0.0551

The corresponding learning curves are shown in Figure 1 and Figure 2. For the cost regime, the learning curves show no clear trend and resemble noise. This reinforces the observation that all configurations score similarly, and that the differences likely reflect evaluation noise rather than real performance gaps. It further raises the question of whether the agent learns a meaningful policy at all in this regime, as the reward does not visibly increase over the training period.

For the no-cost regime, a clear trend, plateau, and convergence can be seen. The mean validation reward increases until roughly 0.5×10^6 training steps, after which it plateaus and remains stable at approximately -0.012 for the remainder of

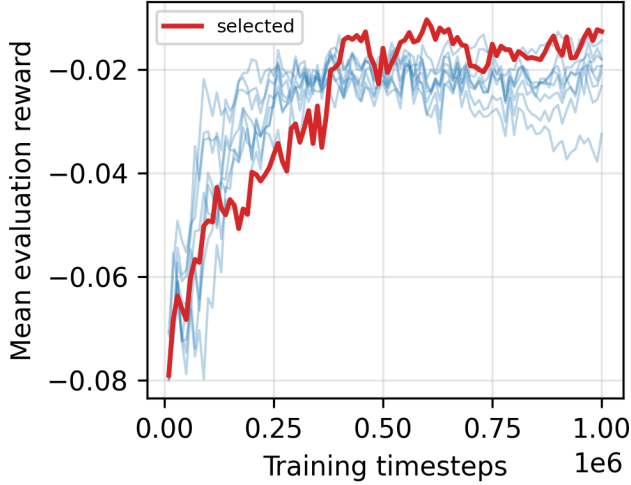


Fig. 1: Mean validation reward over 100 held-out episodes during training, no-cost regime ($\beta = 0$). The selected configuration (3×10^{-4} , $\epsilon = 0.2$) is highlighted.

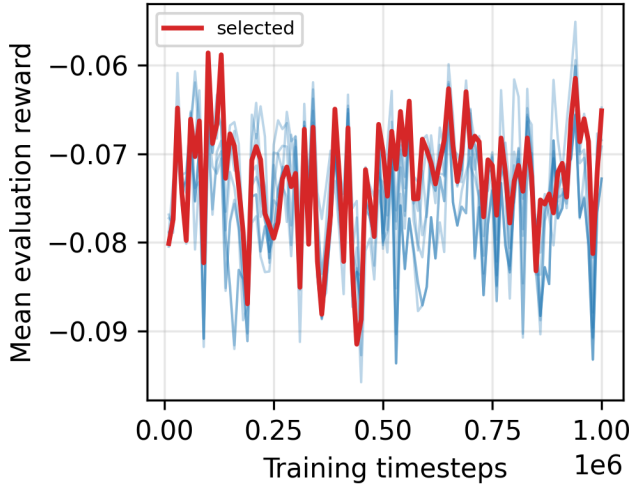


Fig. 2: Mean validation reward over 100 held-out episodes during training, cost regime ($\beta = 0.005$). The selected configuration (3×10^{-4} , $\epsilon = 0.2$) is highlighted.

training. The selected configuration performs noticeably better than most others, although several other configurations seem to converge over a similar number of training steps.

B. Hedging Without Transaction Costs

The portfolio results of the model trained in the no-cost regime are presented in Table IV. From the table it can be observed that the mean portfolio values of both policies at T are fairly close, with a paired mean difference of only -0.004 . The paired difference statistics indicate that the means are statistically indistinguishable, as $p = 0.92$, and the confidence intervals of the two policies overlap. This confirms

a similar hedging performance for both policies under the no-cost regime. However, there is a substantial difference in the standard deviation of the terminal portfolio value, and thus the minimum and maximum terminal portfolio values. This observation is reinforced by the histogram plotting the distribution of the terminal portfolio values $\Pi(T)$ seen in Figure 3. For the BS Delta policy an approximately Gaussian shape centered around its mean is observed, while for the no-cost policy this is not the case. The distribution is wider and somewhat heavier in the left tail, ranging from -6.10 to 3.87 against $[-2.67, 2.31]$ for the benchmark. This seems to indicate that although the final performance of the no-cost policy can be considered equal, the minimum portfolio value, but also the maximum value, are further apart.

TABLE IV: Terminal portfolio value $\Pi(T)$ statistics, no-cost regime ($\beta = 0$) at $K = 100$, over $m = 1000$ paired evaluation paths. MAE denotes the mean absolute error.

Strategy	Mean	95% CI	Std	MAE	Min/Max
PPO	-0.027	± 0.099	1.591	1.289	$-6.10 / 3.87$
BS Delta	-0.022	± 0.039	0.624	0.468	$-2.67 / 2.31$
Paired difference (PPO – BS): mean -0.004 , $t = -0.10$, $p = 0.92$					

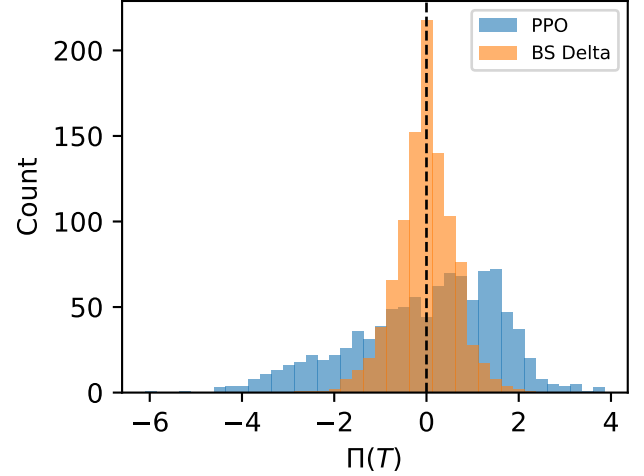


Fig. 3: Distribution of terminal portfolio value $\Pi(T)$ for the PPO agent and Black-Scholes delta hedging, no-cost regime ($\beta = 0$) at $K = 100$, over $m = 1000$ paired evaluation paths.

Figures 4, 5, and 6 illustrate the learned hedging behavior on out-of-the-money, at-the-money, and in-the-money paths. For all three different scenarios the no-cost policy follows the theoretical delta $\Delta(t)$ closely. The most accurate tracking is that for the ITM path. The OTM path seems to have some significant differences from the theoretical delta after $t/N \approx 0.50$, which could be explained by a lack of training samples. This is due to the GBM drift parameter μ in Equation 18 being a positive value, meaning the asset paths drift to the ITM region over time. Additionally, part of the difference can

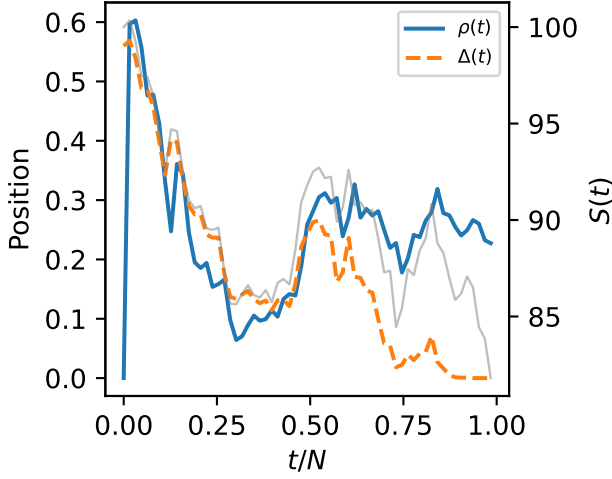


Fig. 4: Learned position $\rho(t)$ and Black-Scholes delta $\Delta(t)$ (left axis) with the underlying price $S(t)$ (right axis, grey), no-cost regime, out-of-the-money path ($S_T/K = 0.83$).

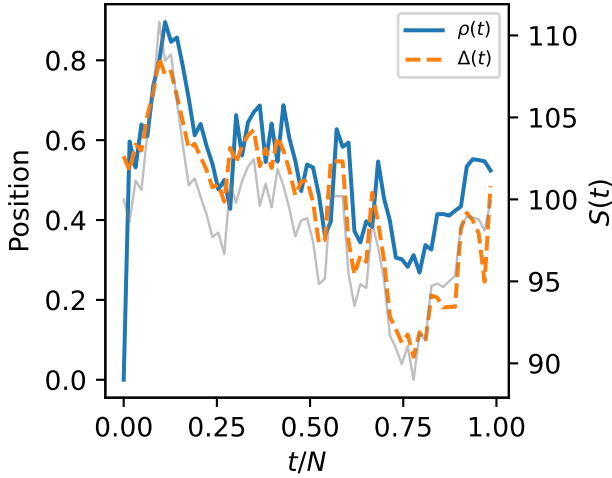


Fig. 5: Learned position $\rho(t)$ and Black-Scholes delta $\Delta(t)$ (left axis) with the underlying price $S(t)$ (right axis, grey), no-cost regime, at-the-money path ($S_T/K = 1.00$).

be explained by the agent only rebalancing after the stock price, and thus the theoretically correct delta, has moved. This is true for all scenarios, and this inherently will produce an imperfect hedge due to a one-step information lag. However, coming back to the final results, it can be stated that over a sufficient number of runs this discrepancy only affects the standard deviation of the terminal portfolio values, and not the mean.

C. Hedging With Transaction Costs

For the cost regime the same experiment is performed as for the no-cost regime, of which the results are presented in

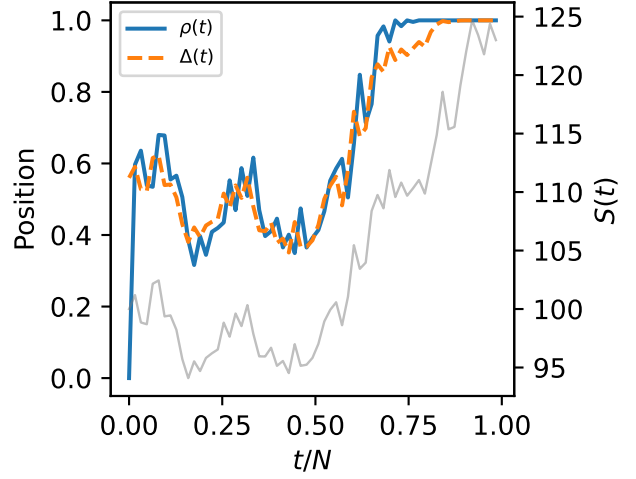


Fig. 6: Learned position $\rho(t)$ and Black-Scholes delta $\Delta(t)$ (left axis) with the underlying price $S(t)$ (right axis, grey), no-cost regime, in-the-money path ($S_T/K = 1.22$).

Table V. In this case, the mean terminal portfolio value of the cost policy (-1.014) is notably higher, i.e., less negative than that of the Black-Scholes policy (-1.835), with a difference of 0.821 which the paired test confirms to be statistically significant ($t = 3.16$, $p = 0.002$). At first glance, this suggests the agent outperforms the benchmark; however, examining the transaction costs reveals the true cause. The PPO policy incurs a mean cumulative transaction cost of only 0.796 , against 1.802 for the Black-Scholes policy. This is a saving of roughly 1.0 . This cost saving more than accounts for the 0.821 difference in mean portfolio value, so the agent's higher mean is driven almost entirely by its trading less, not by hedging more effectively.

TABLE V: Terminal portfolio value $\Pi(T)$ statistics, cost regime ($\beta = 0.005$) at $K = 100$, over $m = 1000$ paired evaluation paths. Cost denotes mean cumulative transaction costs.

Strategy	Mean	95% CI	Std	MAE	Cost	Min/Max
PPO	-1.014	± 0.513	8.277	6.780	0.796	$-36.20 / 10.71$
BS Delta	-1.835	± 0.053	0.855	1.835	1.802	$-5.29 / 0.08$
Paired difference (PPO – BS): mean 0.821 , $t = 3.16$, $p = 0.002$						

That the cost policy's advantage comes with a trade-off becomes clear from the standard deviation. The PPO policy has a standard deviation of 8.277 against only 0.855 for the benchmark, which is roughly ten times larger, with a far heavier left tail and a MAE of 6.780 against 1.835 . The cost policy has a minimum of -36.20 , against a minimum of -5.29 for the Black-Scholes policy. The distribution of $\Pi(T)$ in Figure 7 confirms this, as the Black-Scholes policy remains narrowly concentrated near its mean, while the PPO policy is spread widely. The agent thus achieves a marginally higher

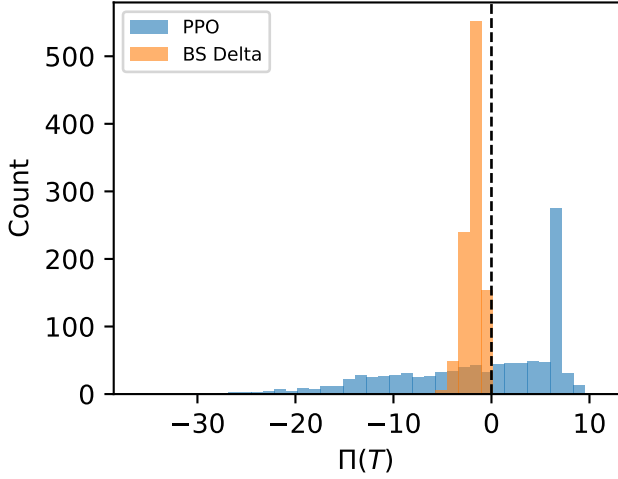


Fig. 7: Distribution of terminal portfolio value $\Pi(T)$ for the PPO agent and Black-Scholes delta hedging, cost regime ($\beta = 0.005$) at $K = 100$, over $m = 1000$ paired evaluation paths.

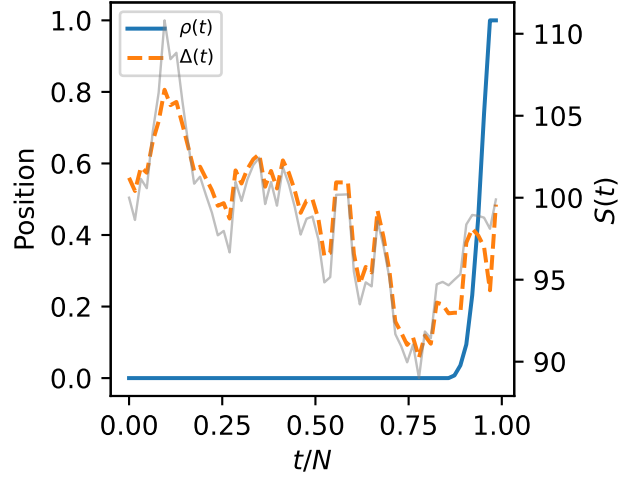


Fig. 9: Learned position $\rho(t)$ and Black-Scholes delta $\Delta(t)$ (left axis) with the underlying price $S(t)$ (right axis, grey), cost regime, at-the-money path ($S_T/K = 1.00$).

mean only by avoiding transaction costs, at the expense of drastically increasing the potential losses of the portfolio.

Figures 8, 9, and 10 explain this behavior. In contrast to the no-cost regime, the cost policy no longer tracks the theoretical delta $\Delta(t)$. Instead, it acquires a position and holds it largely unchanged, rebalancing almost never. This near-static hedge minimizes the transaction costs incurred over the option's life, which the reward penalizes at every step, but leaves the portfolio poorly hedged against the underlying asset movements. The agent has effectively learned that, under proportional transaction costs, continuous rebalancing toward the theoretical delta is not worth the transaction costs it incurs.

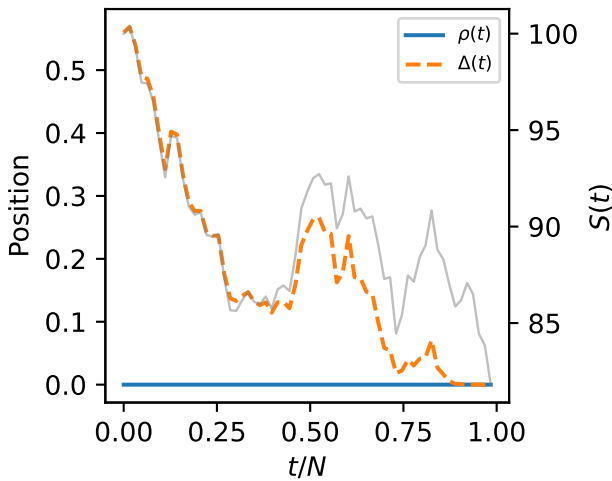


Fig. 8: Learned position $\rho(t)$ and Black-Scholes delta $\Delta(t)$ (left axis) with the underlying price $S(t)$ (right axis, grey), cost regime, out-of-the-money path ($S_T/K = 0.83$).

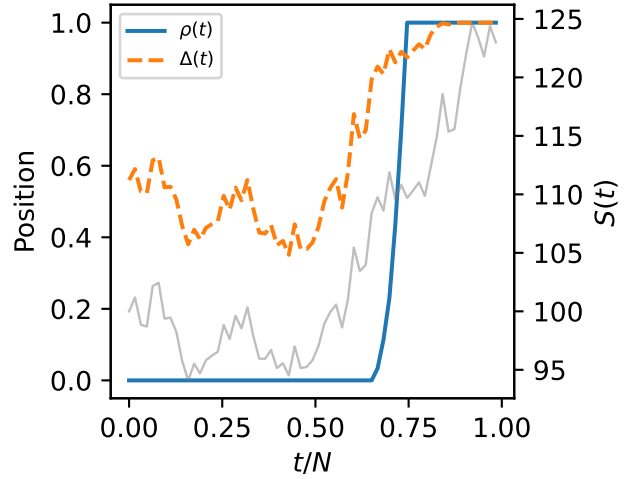


Fig. 10: Learned position $\rho(t)$ and Black-Scholes delta $\Delta(t)$ (left axis) with the underlying price $S(t)$ (right axis, grey), cost regime, in-the-money path ($S_T/K = 1.22$).

D. Comparison of Cost Regimes

To compare the target position $\rho(t)$ selected by each policy against the theoretical Black-Scholes delta $\Delta(t)$, a target-position surface is plotted for each policy, queried from a flat holding. Figure 11 shows the Black-Scholes surface as a function of time-to-maturity and moneyness. When the option is at the money, $\Delta(t) \approx 0.5$; for deep in- or out-of-the-money states it approaches 1 and 0 respectively. As maturity approaches, the transition from 0 to 1 around $S/K = 1$ sharpens, since the underlying has progressively less time to move across the strike.

TABLE VI: Standard deviation of $\Pi(T)$ by strike level K , for the PPO agent and Black-Scholes delta hedging, in both regimes, over $m = 1000$ paths per strike. The agent is trained only at $K = 100$.

K	$\beta = 0$		$\beta = 0.005$	
	PPO	BS Delta	PPO	BS Delta
80	0.841	0.253	9.107	0.390
85	1.141	0.379	9.661	0.531
90	1.375	0.482	9.697	0.637
95	1.517	0.571	9.217	0.752
100	1.591	0.624	8.277	0.855
105	1.683	0.649	7.067	0.936
110	1.851	0.610	5.820	0.959
115	1.935	0.547	4.623	0.951
120	1.908	0.513	3.611	0.933

Figure 12 shows the surface for the no-cost policy, which closely matches the theoretical surface, confirming that the agent has approximately recovered the Black-Scholes delta without ever observing it. The main difference is that near maturity the target position does not fully converge to zero for low moneyness, indicating the agent retains a small residual hedge where theoretically no hedge is needed.

Figure 13 shows the surface for the cost policy, where the target position is near zero across most of the domain. Only close to maturity, and when the contract is clearly in the money, does it deviate, and even then only to roughly 0.4–0.5 rather than to one. The agent has learned that hedging is not worth its transaction cost until the payoff can be predicted with greater certainty. Although this cost policy attains a mean $\Pi(T)$ closer to zero than the Black-Scholes policy in the cost regime (Table V), this reflects its lower transaction costs rather than better hedging. The standard deviation of $\Pi(T)$ is higher than that of both the no-cost and Black-Scholes policies, indicating substantially worse replication.

Table VI reports the standard deviation of $\Pi(T)$ across strike levels for both regimes, with the agent trained only at the at-the-money contract ($K = 100$). In the no-cost regime, the agent’s standard deviation increases with strike, from 0.84 at $K = 80$ to 1.91 at $K = 120$. The Black-Scholes benchmark exhibits a different shape, peaking near the money (0.65 at $K = 105$) and decreasing towards both wings, reflecting that discrete-hedging error is largest where Gamma is highest. The agent’s error does not diverge at strikes far from $K = 100$: it remains within a factor of roughly 2.5–3.7 of the benchmark across the entire range, and the cumulative hedging error ϵ_{cum} stays within a narrow band (3.1–6.4). The policy therefore generalizes to strike levels unseen during training, supported by the moneyness-normalized state space of Eq. (7), though its tracking is relatively weakest for high-strike out-of-the-money options, which are least represented in at-the-money training paths.

In the cost regime, the agent’s standard deviation is substantially larger than the benchmark at every strike (for example 9.11 versus 0.39 at $K = 80$), and the cumulative hedging error is correspondingly large (ϵ_{cum} of roughly 7–32 versus 3–6 in

the no-cost case). Whereas the benchmark’s standard deviation remains low and stable across all strikes (0.39–0.96), the agent’s is several times higher, confirming that this degradation stems from the agent’s near-static policy rather than from a failure to generalize across strikes. The agent’s cost-regime error follows the opposite trend to the no-cost case, decreasing with strike: for low strikes the option is likely to finish in the money, so the failure to hedge dynamically leaves the largest absolute exposure, whereas for high strikes the option usually expires worthless and minimal hedging suffices.

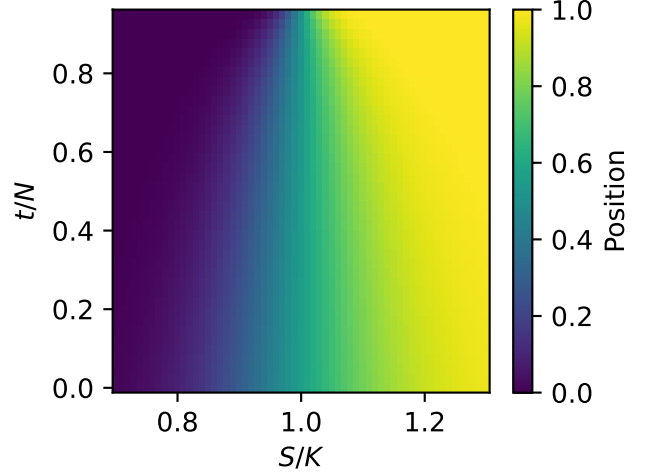


Fig. 11: Target position selected by the Black-Scholes policy as a function of moneyness S/K and elapsed time t/N , queried from a flat holding ($\rho(t-1) = 0$).

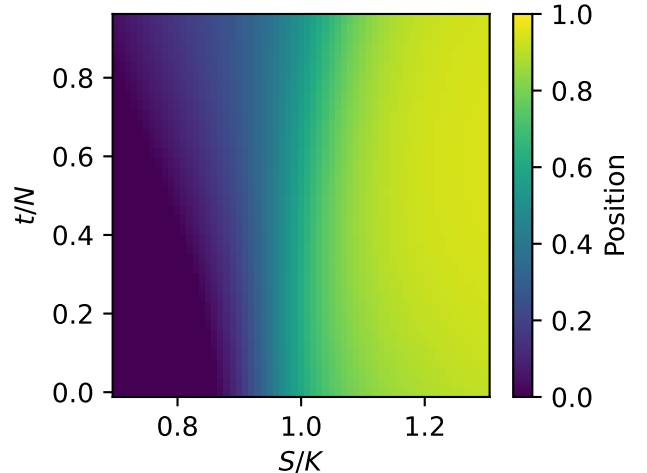


Fig. 12: Target position selected by the no-cost policy as a function of moneyness S/K and elapsed time t/N , queried from a flat holding ($\rho(t-1) = 0$).

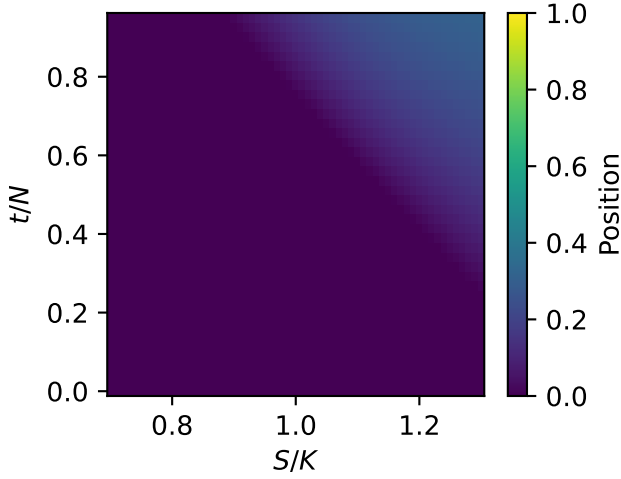


Fig. 13: Target position selected by the cost policy as a function of moneyness S/K and elapsed time t/N , queried from a flat holding ($\rho(t-1) = 0$).

E. Robustness to Volatility Mismatch

To assess robustness to out-of-training volatility regimes, both policies were evaluated on paths simulated under a range of volatilities σ , while hedging under the training volatility $\sigma = 0.30$. The results are reported in Table VII. In both cost regimes the standard deviation of $\Pi(T)$ increases as the evaluation volatility rises, which is expected: higher volatility permits larger moves in the underlying asset, making the option harder to hedge. Notably, the Black-Scholes benchmark attains its minimum standard deviation at $\sigma = 0.30$ and increases for volatilities both above and below it. This U-shaped pattern is direct evidence that the mismatch between the hedging volatility and the realized volatility degrades performance, distinct from the general increase in hedging difficulty at high volatility.

TABLE VII: Standard deviation of $\Pi(T)$ by evaluation volatility σ , for the PPO agent and Black-Scholes delta hedging, in both regimes, over $m = 1000$ paths per level at $K = 100$. Both strategies hedge under the training volatility $\sigma = 0.30$; paths are simulated at the evaluation volatility.

σ	$\beta = 0$		$\beta = 0.005$	
	PPO	BS Delta	PPO	BS Delta
0.2	1.239	0.722	5.755	0.622
0.3	1.591	0.624	8.277	0.855
0.4	2.288	1.231	10.588	1.747
0.5	3.247	2.211	12.795	2.878
0.6	4.381	3.344	14.945	4.120
0.7	5.611	4.566	17.072	5.429
0.8	6.906	5.841	19.085	6.776

In the no-cost regime, the Black-Scholes benchmark attains a lower standard deviation than the agent at every volatility level, but the two converge as volatility rises (Figure 14). The

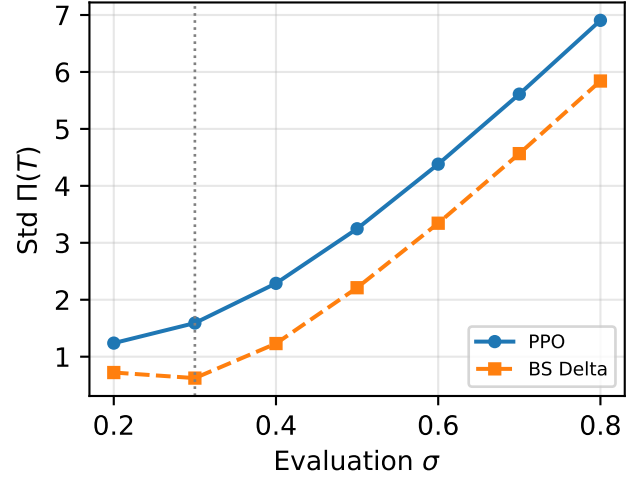


Fig. 14: Standard deviation of terminal portfolio value $\Pi(T)$ for the no-cost policy (trained at $\sigma = 0.30$) hedging under different evaluation volatilities.

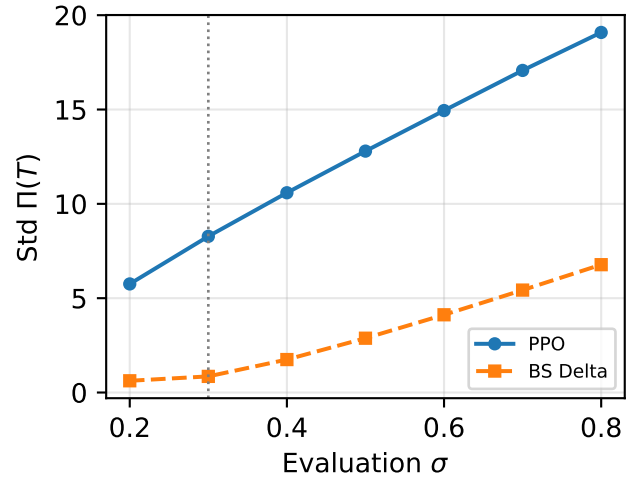


Fig. 15: Standard deviation of terminal portfolio value $\Pi(T)$ for the cost policy (trained at $\sigma = 0.30$) hedging under different evaluation volatilities.

agent's standard deviation is roughly 2.5 times the benchmark at $\sigma = 0.30$, narrowing to about 1.2 times at $\sigma = 0.80$. The agent's error therefore grows more slowly under increasing volatility mismatch than the benchmark's, indicating that its policy is relatively robust to different volatility regimes.

In the cost regime (Figure 15), the agent's standard deviation is significantly higher than the benchmark for all volatility levels, reflecting the near-static hedging policy identified earlier. The same relative convergence is observed as the agent's disadvantage falls from roughly a factor of nine at $\sigma = 0.2$ to a factor of three at $\sigma = 0.8$. The absolute gap nonetheless widens over this range, so in absolute terms the cost agent's

hedging error remains far above the benchmark across all volatility regimes.

VI. DISCUSSION

In this section the practical implications will be discussed, after which the limitations of this study will be highlighted. Building on these limitations, some suggestions will be proposed for future studies.

A. Practical Implications

This paper shows that reinforcement learning, in the absence of transaction costs, can find a policy similar to the theoretical Black-Scholes delta and can attain a similar mean hedging performance. For financial engineers in the industry this means that reinforcement learning can be used to retrieve hedge ratios without relying on a closed-form solution: the method requires only the ability to simulate or observe price paths. This is something the Black-Scholes approach is not able to offer, as it relies heavily on its modeling assumptions. The practical value of this result is therefore not to replace the delta hedge where it is already available in closed form, but to validate the reinforcement learning approach in a setting where the optimal policy is known, before applying it to settings where it is not.

The results under transaction costs carry a more cautionary implication. The agent attained a higher mean terminal portfolio value than the Black-Scholes benchmark, but with a roughly tenfold higher standard deviation, by hedging only rarely. This shows that the reward function effectively defines what good hedging means. The agent optimized the specified objective, in which the cost penalty accumulated at every step while the hedging error was penalized only once at maturity. Because of this reward function the agent sacrificed replication quality for transaction cost savings, resulting in a higher mean portfolio value at maturity. For practitioners this implies that reward design should include the desired risk-preference, as a trading desk that cares about tail risk or variance would find the learned policy unusable. Deployment of reinforcement learning for hedging must include the institution’s actual risk tolerance and policies directly into the objective. Although from a risk-management standpoint the policy is not usable in practice, it does show interesting results in discovering the transaction costs and trading frequency trade-off.

In this study the agent is trained using a state space and reward function normalized by the strike price K . The goal of this normalization was to be able to generalize the policy to different strike prices, while only having trained on simulations for one strike level. From the results in Section V-D it can be concluded that the learned policy indeed generalizes across strikes, with hedging error remaining within a small region of the benchmark over the full evaluated range. Performance does degrade for high-strike OTM contracts, which is explained by these regions being underrepresented in the training paths rather than by a failure of the normalization itself. For practical deployment this is an attractive property, as a single trained model, combined with an appropriately normalized state space, could cover a range of contracts rather than

requiring one model per strike. Additionally, the robustness results in Section V-E show that the agent’s hedging error grows more slowly than the benchmark’s when the realized volatility deviates from the training volatility. Since any hedger must commit to a volatility assumption that will in practice be wrong, a policy that degrades predictably under such a volatility mismatch is desirable.

It should be noted that these practical implications hold within a simulated environment subjected to the GBM assumption, with proportional transaction costs and discretized timesteps. The simplifications made, and the steps required to generalize to real financial markets, are covered in the Limitations (Section VI-B) and Future Work (Section VI-C).

B. Limitations

In this paper we address two of the main simplifying assumptions of the Black-Scholes model for hedging a call option, namely continuous rebalancing and the absence of transaction costs. However, there are limitations to this study itself, as well as simplifications that are not addressed.

The first limitation is regarding the design of the reward function. The transaction cost penalty is incurred at every rebalancing step, while the hedging error is penalized only once, at maturity, through the absolute deviation of the terminal portfolio value. This objective includes no explicit preference regarding variance or tail risk, which is reflected in the policy learned in the cost regime. The learned behavior is a consequence of the chosen objective rather than a general property of reinforcement learning for hedging. Related to this, the transaction cost level $\beta = 0.005$ was chosen to illustrate the effect, and only this single level was tested. Thus it remains unknown how the learned policy changes as β decreases, and whether the transition from delta tracking and portfolio replication to more static hedging occurs gradually or abruptly.

The second limitation is that all training was performed with a single fixed random seed. Policy gradient methods such as PPO are known to be sensitive to initialization, and the sensitivity analysis in this study compared hyperparameter configurations but not the variance across independent training runs. Especially given that the cost-regime learning curves resemble noise, it cannot be ruled out that a different seed would produce a different policy in this regime.

The third limitation is the assumption of a fully liquid market with no slippage costs. In reality, orders are subjected to slippage costs due to order fill times and the possible lack of liquidity of the asset of interest. This study removes both of these frictions by assuming a fully liquid market with no slippage.

Furthermore, this paper considers a European call option with a maturity of 3 months, hedged from inception. This is a relatively simple financial contract, as in the modern financial markets there are many different types of options. Examples are forward-start options, cliquet options, barrier options, Asian options, and American options. Additionally, the option is assumed to be sold and hedged from the start of the contract. In reality, a hedging desk may take over or

initiate a hedge partway through a contract’s life, with less time remaining before the option expires.

The final limitation is in the asset price simulation. For the purpose of this research Geometric Brownian Motion was assumed, with a given fixed risk-free rate r and volatility σ . In reality, both the risk-free rate and volatility are not fixed or given, but rather change over time. Especially volatility is uncertain, even over short time periods. Stochastic volatility models like the Heston model [22] are better able to capture real volatility behavior. For the risk-free rate, a mean-reverting process such as the Ornstein-Uhlenbeck-based Vasicek model [23] could give more realistic behavior. Additionally, the diffusion process uses a Wiener process, implying normally distributed log-returns, while empirical evidence suggests that real return distributions have fatter tails [24]. Jump-diffusion models could capture such behavior more accurately. Note that even the robustness experiments in this study vary parameters within the GBM framework. This means the agent was never evaluated on paths generated by a different process, so its performance under more realistic price dynamics remains undiscovered.

C. Future Work

As addressed in the limitations, there are several simplifications and design choices that could be revisited in future research. The first concerns the reward function. Future work could incorporate an explicit risk parameter into the objective, for example through a variance penalty, a risk measure such as CVaR as used in the deep hedging framework [19], or a per-step hedging error signal that decreases the dependence on the single terminal reward. Related to this, a sweep over the transaction cost level β could reveal whether the transition from delta-tracking to near-static hedging happens gradually or abruptly, and training over multiple random seeds would establish whether the learned cost-regime policy is a stable outcome or an artifact of a single run.

The second direction is to include, or center around, different option types with path-dependent payoffs. Whereas the European call studied here depends only on the terminal asset price, contracts such as Asian, barrier, and cliquet options depend on the full price path. This poses an extra challenge for the agent, as the state space would need to be extended with path information, and it raises the question of whether strike generalization would still hold for such contracts. Increasing the time to maturity could also prove useful, as in reality option contracts come in many different durations.

The third proposal for future work is the inclusion of different asset price dynamics, such as stochastic or local volatility models, or a model using a jump-diffusion process. These types of models are better able to capture the behavior of real asset prices, and thus using these as building blocks in the environment simulation improves the applicability of the learned policy. Additionally, testing on real market data could yield useful insights into the behavior of the policy in real market environments.

The last recommendation for future research is the effect of “netting” in hedging and portfolio management. This study focuses on a portfolio consisting of only one asset and one option contract. In reality, financial institutions hold many of these types of assets on their books, and constantly buy and sell these, effectively altering their risk exposure. Hedging all individual positions is then almost impossible, and these institutions can leverage the netting effect, whereby positions partially offset each other’s risk, removing the need to actively hedge each contract individually. Studying a reinforcement learning agent hedging a portfolio of several option contracts with different strikes and maturities could therefore be useful for practitioners in the financial industry.

VII. CONCLUSION

This paper studied the performance of a PPO reinforcement learning agent in hedging a vanilla European call option. Under the no-cost regime, the agent produces results similar to the theoretical Black-Scholes approach, closely matching both its mean hedging performance and its delta surface without ever observing the theoretical delta. The policy trained in the cost regime changes compared to the no-cost policy, as it hedges far less frequently. It seems to try and predict the final asset value when closer to maturity and to hedge only when needed, avoiding the transaction costs that would otherwise reduce the terminal portfolio value. The results show that this policy attains a higher mean terminal portfolio value, driven by lower transaction costs rather than better replication. However, it cannot be considered a useful replacement when risk control and management are a concern.

By normalizing the state space and rewards, the agent is able to learn a generalizable hedging policy while only training on at-the-money paths. This approach could prove useful for other types of options, as financial engineers would not have to train a separate model for every strike level, which would require significant computational costs.

Testing the policy on different, unseen volatility regimes does degrade the hedging performance of the agent significantly. However, it does so in a rather predictable way, meaning the degradation can be accounted for when assessing hedging efficiency. To conclude, reinforcement learning shows promising results for hedging derivatives, but whether it can replace delta hedging in practice depends critically on including the appropriate risk preference into the objective.

AI STATEMENT

An AI-assisted tools (Claude by Anthropic) was used in the preparation of this manuscript in the following capacities:

Algorithmic Development: Claude was used to generate code outlines for the process mining implementation and support debugging.

Report Refinement: Claude was used to improve the academic tone of the report, refine mathematical formulations, and correct grammar, spelling, and ensure consistency.

The author takes full responsibility for all technical content, results, and conclusions presented in this work.

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